

PRESS RECORD

COMPX341 - Advanced Software Engineering Design



THE UNIVERSITY OF
WAIKATO
Te Whare Wānanga o Waikato

Software Maintenance – an introduction

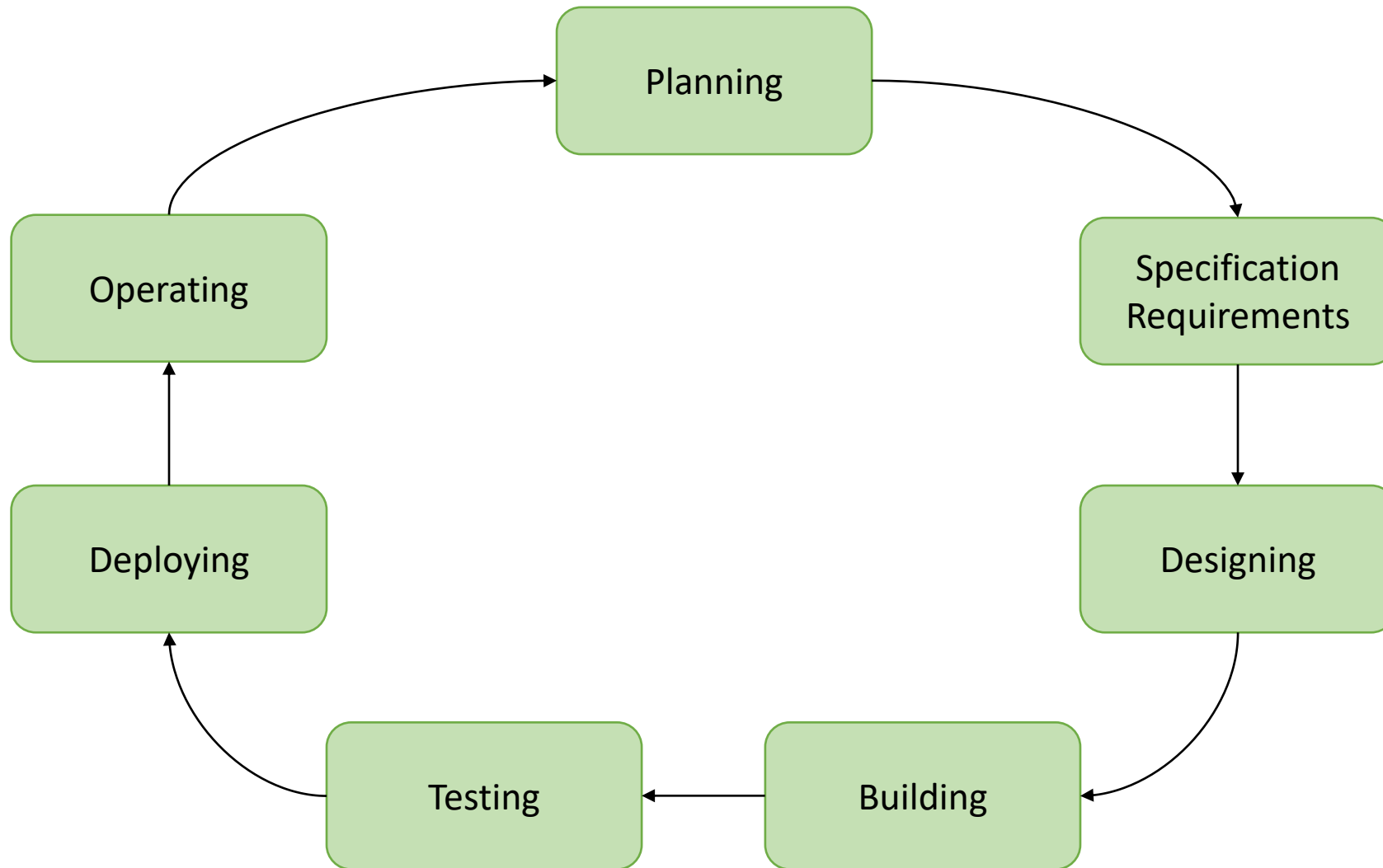
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Overview

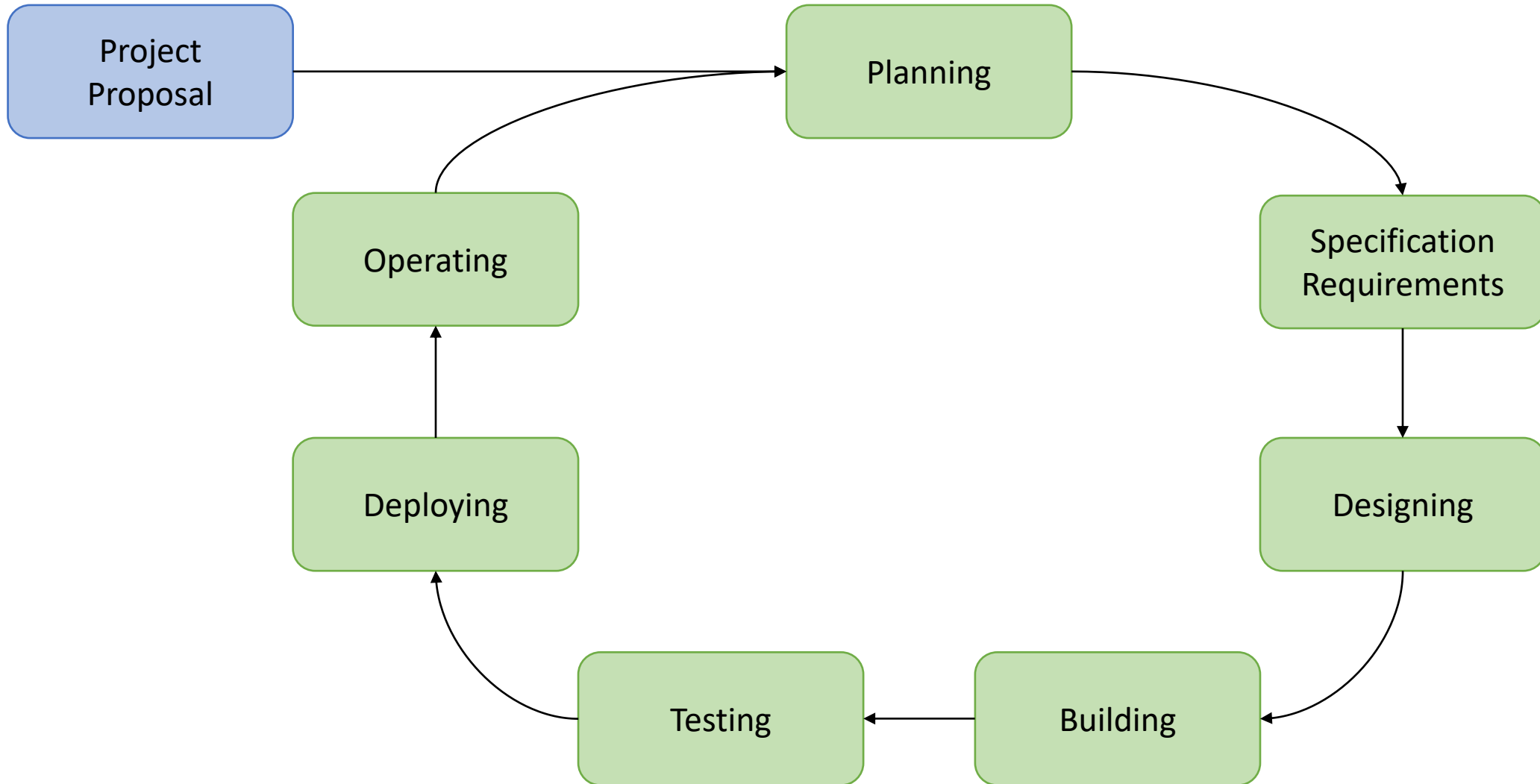
- Software lifecycle
- Software maintenance
- Self-adaptive Systems

Software Life Cycle

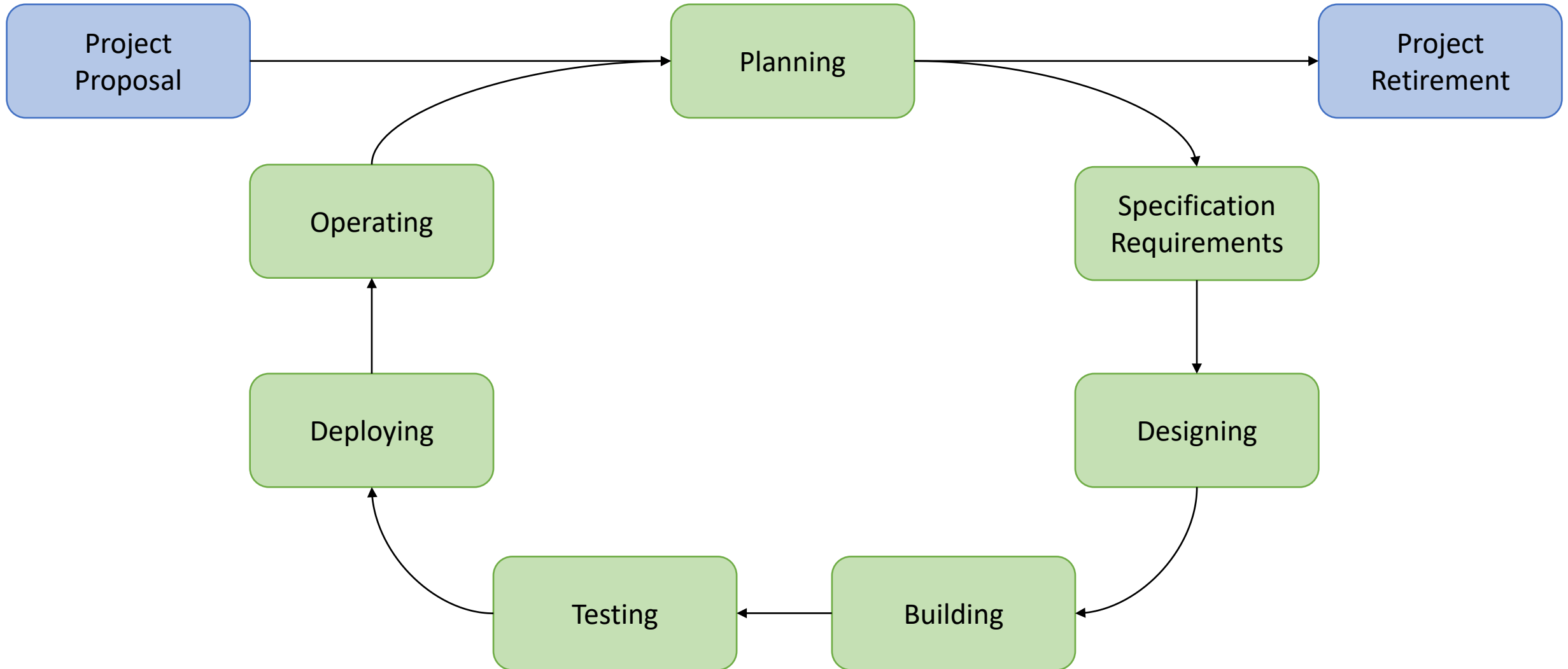
Software Life Cycle



Software Life Cycle



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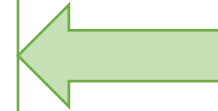


Software Life Cycle

- > Project proposal
- > 1st iteration (planning, requirements, ..., deployment)
- > Operation
- > 2nd iteration (planning, requirements, ..., deployment)
- > Operation
- ...
- > Nth iteration (planning, requirements, ..., deployment)
- > Operation
- > Retirement

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Software Maintenance
describes these parts
of the Software Life Cycle

Software Maintenance

Software Maintenance

“... all the activities required to provide cost-effective support to software”

-- SWEBOK v3

“The objective of Software Maintenance is to modify existing software while preserving its integrity”

-- SWEBOK v3

- Essentially, modifications after delivery
- Conducted by maintainer

Need for Software Maintenance

1. Correct faults
2. Improve the design
3. Implement enhancements
4. Interface with other software
5. Adapt programs so that different hardware, software, system features, and telecommunication facilities can be used
6. Migrate legacy software
7. Retire software
8. Maintaining control over the software's day-to-day functions
9. Maintaining control over software modification
10. Perfecting existing functions
11. Identifying security threats and fixing security vulnerabilities
12. Preventing software performance from degrading to unacceptable levels

Reasons for Maintenance

- Correction
 - Same requirements
 - E.g. Correcting an existing feature based on existing requirements
- Enhancement
 - Changes requirements and/or changed environment
 - E.g. Adding a new feature before of changes to the requirments

Times for Maintenance

- Proactive
 - To prevent failure
 - E.g. making the change prior to failure occurring
 - Failure can refer to feature failure, system failure, or failure to meet requirements
- Reactive
 - Because of one or more reported failures
 - E.g. making the change after a failure has occurred (and been reported)
 - Failure can refer to feature failure, system failure, or failure to meet requirements

Software Maintenance Categories

- Preventative
- Perfective
- Corrective
- Adaptive

Software Maintenance Categories

	Correction Same requirements	Enhancement Changes requirements and/or changed environment
Proactive To prevent failure	Preventive Bugs are there but not surfacing ... yet!	Perfective New features or documentation; better performance; etc.
Reactive Because of one or more reported failures	Corrective a.k.a. Repairs Special case: emergency	Adaptive Changed/changing environment (e.g., updated OS, middleware, etc.)

Software Maintenance Categories

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Examples: what kind of maintenance is required?

Issue description	Preventative Proactive- correction	Perfective Proactive- enhancement	Corrective Reactive- correction	Adaptive Reactive- enhancement
One of our testers realised the deployed product could crash if a user logs in from two devices at the same time				
It's been recommended we add a bar chart plot to this report				
The web app is getting slower and slower now that we've increased our client base so much!				
The client called as at 2am – one hour ago. “I have no clue why all these train tickets were sold at half the price.”				

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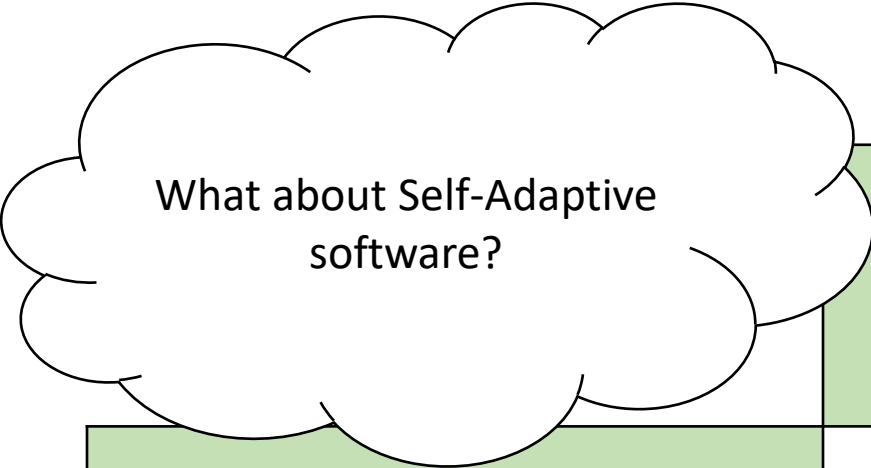
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- Software that tries to maintain itself!
 - i.e. self-configuration: changes configuration constants
 - i.e. self-optimization: changes optimization parameters
 - i.e. self-protection module: filters external attacks
 - i.e. self-healing: ready-made mechanisms to ensure the system continues operating as intended
- E.g. auto-scaling automatically looks at and adapts the system such that it has the ideal number of instances
- E.g. cloud often has the capacity to bring up an application that has crashed (self-healing)

What is self-adaptive software?

- Note: the maintenance team is still there!
- Self-adaptive systems just try to minimise their involvement

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